
Evaluating Predictive Gait Representations Beyond Latent-Feature Matching

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Abstract

1 Foundation models for temporal systems are often judged by how well they satisfy
2 a pretraining objective, but that score may not show whether the learned represen-
3 tation supports a temporal quantity needed downstream. We study this gap in a
4 controlled skeleton-sequence setting. A small S-JEPA learns to predict hidden
5 pose features, while a frozen ridge readout estimates a timestamp-derived contrast
6 between left- and right-side pose-coordinate speeds in gait clips. We compare
7 each trained teacher with its matched initialization under source-held-out evalu-
8 ation. Training makes hidden-feature predictions favor the matching clip more
9 consistently, yet reduces movement-readout accuracy across masking strategies
10 and seeds. The experiment does not test future-only forecasting or foundation-
11 model scale. Instead, this negative result motivates task-aligned, leakage-aware
12 evaluation of temporal foundation models, with pretraining diagnostics reported
13 separately from the system measurements that a downstream task requires.

14 1 Introduction

15 Masked prediction lets temporal models learn from sequences without behavior labels. S-JEPA
16 predicts teacher features for hidden parts of a sequence from visible context. That objective measures
17 feature matching, not whether a later readout can estimate a chosen movement quantity. We test that
18 distinction in gait.

19 Walking must be measured over time because a single pose does not show how body landmarks
20 move. Pose sequences remain incomplete: camera angle and estimation error affect the coordinates,
21 while forces and foot contact are unobserved. We study a narrow quantity available from the poses
22 themselves, a signed contrast between left- and right-side coordinate speeds. The score is not a
23 clinical measure of gait asymmetry.

24 We compare two evaluations of the same S-JEPA models [1]. The first asks whether predictions
25 from visible pose patches are closer to hidden teacher features from the same clip than to features
26 from another video. The second freezes the encoder and uses ridge regression to predict the left-right
27 speed score. Each trained encoder is paired with its own initialization, and whole source videos are
28 held out. After training, every recorded diagnostic comparison favors the matching clip, yet mean
29 R^2 for the speed score falls by 0.108-0.122 across mask conditions.

30 Masked skeleton models are usually judged by action recognition [1] [2] [3]. Video anticipation and
31 predictive world models instead assess future behavior [4] [5]. Our study uses S-JEPA as a controlled
32 case and tests a continuous movement readout before and after training. The finding is specific to this
33 dataset, model, target, and readout, but it shows why feature matching and downstream measurement
34 should be evaluated separately.

2 Study design

2.1 Left-right speed score

After quality checks, the dataset contains 625 clips from 93 GAVD source videos [6]. Each clip provides 33 pose landmarks with three estimated coordinates, visibility, frame numbers, and frame rate. The coordinates use image normalization and inferred depth rather than calibrated physical units. For the target calculation, we keep the original timestamps and gaps, center each observed pose at the pelvis, and divide the coordinates by body width.

We use five left-right landmark pairs: shoulders, knees, ankles, heels, and foot tips. Each pair must have at least eight transitions for which both sides are observed at both endpoints. For each valid transition, we divide a landmark’s Euclidean displacement in the three normalized coordinates by the elapsed time. Let $m_{L,k}$ and $m_{R,k}$ be the median left- and right-side speeds for pair k . We define

$$y = \frac{1}{5} \sum_{k=1}^5 \frac{m_{L,k} - m_{R,k}}{m_{L,k} + m_{R,k} + 10^{-8}}.$$

Positive scores mean that the left-side landmarks have higher median pose-coordinate speeds on average; negative scores mean the opposite. Anatomical reflection changes the sign. Reversing the order of a sampled trajectory and its time intervals leaves the score unchanged because the calculation uses speed magnitudes. The score therefore does not test temporal direction, gait phase, stride timing, or loading.

2.2 S-JEPA training

The model input follows a separate preparation path (Figure 1). It fills interior gaps of up to four samples, centers coordinates at the pelvis, scales them by body width, and resamples each clip to 64 equally spaced sequence positions. When the pelvis is missing, this path uses a fallback estimate; unlike the target path, it does not give the encoder original elapsed times or clip duration. Four positions from one joint form a patch, producing 16×33 tokens; invalid patches are masked out. Appendix A gives the visibility rules and tensor dimensions.

S-JEPA uses visible patches to predict teacher features at hidden positions. The teacher sees the complete prepared clip, and its weights follow an exponential moving average of the online encoder. A VICReg term [7] also makes pooled features from two geometric views agree while discouraging collapsed or redundant features. Neither the speed score nor gait-category labels enter training.

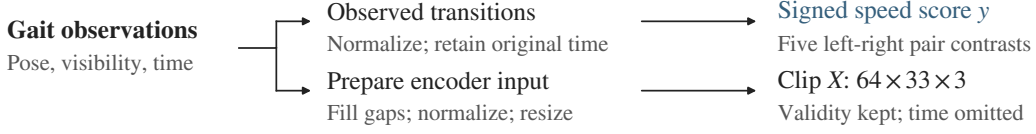
We test five mask conditions. The motion family uses random targets, MAMP-style motion-weighted targets [2], or a mixture of random and motion-weighted targets. The region family samples targets either uniformly or from connected body landmarks. Each structured mask has a count-matched random control with the same initialization, data draws, views, and number of updates. Across five outer folds and five seeds, this gives 125 trained models, each run for 1,200 updates. This grid does not use reflection augmentation or a reflection loss.

2.3 Evaluation against matched initialization

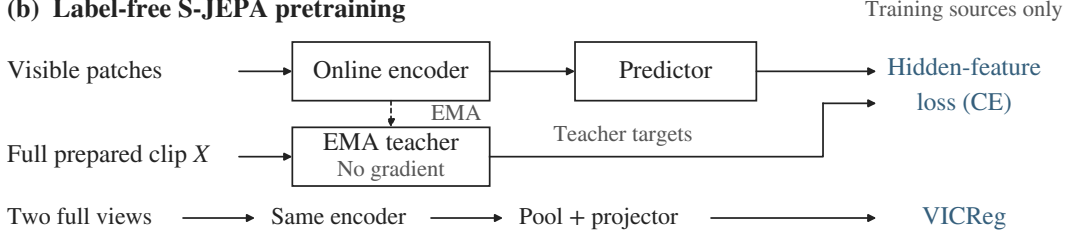
Five outer folds keep all clips and derived versions from a source video together. For each fold, the encoder trains on the other four folds only; we then freeze its initialization and trained weights. Within the outer-training data, three source-separated inner folds choose the ridge penalty. Imputation and scaling are fitted only on each inner-training partition and then refitted on all outer-training sources. No outer-test clip enters encoder training, readout fitting, or model selection. An inner-validation clip may still have contributed without labels to encoder training.

The expanded readout used for the main results has 2,890 inputs: feature means, standard deviations, absolute changes between consecutive blocks, and ten observation-support fractions. For each seed, we pool predictions from all five test folds, give each source video equal total weight, and compute one R^2 before averaging across seeds. The paired 95% intervals were computed from 2,000 source-video resamples while keeping fitted models and splits fixed. They do not include retraining, new splits, or uncertainty from analysis choices.

(a) Separate measurement and model input



(b) Label-free S-JEPA pretraining



(c) Frozen evaluation



Figure 1: **Study workflow.** (a) Timestamped pose transitions define the speed score; prepared encoder inputs omit elapsed times. (b) Solid arrows carry data; the dashed arrow updates teacher weights by exponential moving average (EMA). CE denotes cross-entropy loss. (c) Separate ridge readouts compare frozen initialized encoders and trained teachers on held-out source videos.

3 Results

3.1 Left-right speed prediction falls after training

With the initialized encoder, the ridge readout reaches mean source-balanced $R^2 = 0.223$. Across the five mask conditions, trained teachers average 0.101-0.114, a drop of 0.108-0.122 (Table 1). Each condition is below its matched initialization in all five seeds, and every condition-level interval is below zero. Mean absolute error rises from 0.0415 to 0.0436-0.0440. The final online encoders also have lower mean R^2 than their initial weights in every condition.

Table 1: **Frozen-feature ridge prediction of the left-right speed score.** Values average five seeds; each pools held-out predictions across five folds with equal total weight per source video. ΔR^2 is trained teacher minus matched initialization, computed before rounding. Its paired 95% intervals use 2,000 source-video bootstrap resamples with fitted models and splits fixed.

Training mask	Trained teacher R^2	ΔR^2	95% interval for ΔR^2
Random (motion)	0.114	-0.109	[-0.170, -0.041]
MAMP-style	0.113	-0.110	[-0.165, -0.051]
Motion mixture	0.114	-0.108	[-0.171, -0.042]
Random (region)	0.109	-0.113	[-0.166, -0.060]
Connected region	0.101	-0.122	[-0.183, -0.055]

3.2 Trained models favor the matching clip

The same-clip diagnostic asks a different question. With visible context and hidden positions fixed, we compare prediction error against teacher features from the matching clip and from another source video. Prediction error is lower for the matching clip in all 375 trained comparisons. At initialization, the matching clip has lower error in 33 of 75 unique comparisons; the duplicated initial control shared across the two mask experiments is counted once (Figure 2). Because these comparisons reuse videos and models, they are not independent trials. Shared pose, viewpoint, or missingness can provide the cue, and raw errors cannot be compared across trained models because each has its own teacher space and scale.

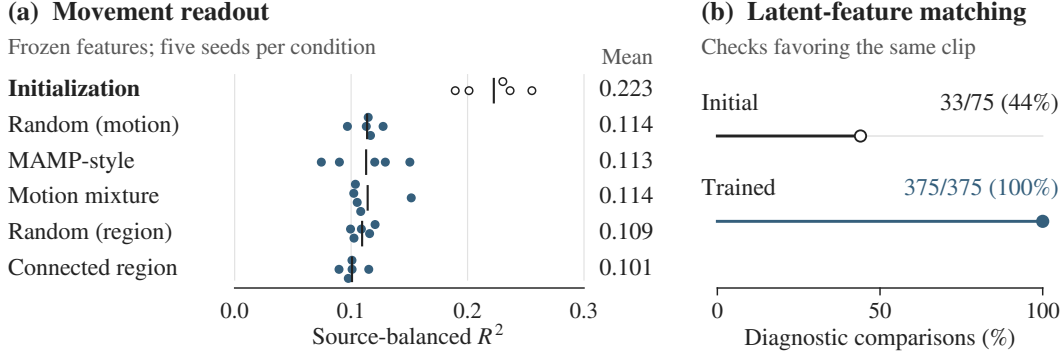


Figure 2: **Speed readout and same-clip matching.** (a) Held-out ridge prediction of the left-right speed score: circles show five seeds’ source-balanced R^2 ; ticks mark means. Open circles denote initialization, filled circles trained teachers. (b) Comparisons favoring the matching clip over another source video by mean squared error. Each aggregates clips per model and mask; shared initializations are counted once. Comparisons reuse videos and models and are not independent trials.

3.3 Secondary checks do not identify a cause

At initialization, the expanded readout raises R^2 from 0.071 for mean features alone to 0.223. It adds variation, feature-change, and observation-support summaries together, so the gain cannot be assigned to temporal order. The baseline also includes a pretrained pose detector, input preparation, joint identities, and supervised ridge regression.

No structured mask has a clear readout advantage over its count-matched random control. The MAMP-style, motion-mixture, and connected-region effects are -0.0008, +0.0005, and -0.009 in R^2 ; all three intervals include positive and negative values. The mask effects remain uncertain.

We also apply the speed formula to the prepared encoder input at uniform relative time steps. On the 623 clips from 92 sources with finite values on both paths, equal-source weighting gives 70.4% sign agreement and $R^2 = 0.218$ between the calculations. No regression is fitted, and no uncertainty interval was computed. Because timing, interpolation, normalization, and valid support all change, this check neither identifies a cause nor sets a performance ceiling.

4 Discussion

The same-clip diagnostic can use any cue that distinguishes one clip from another, including pose, viewpoint, or missingness. The readout instead needs variation aligned with the left-right speed score. Training may strengthen the first set of cues without helping the second. We cannot identify the mechanism, but the two metrics should not be treated as substitutes.

The tested ridge readout may miss information available to another model or feature summary, and its largest penalty is selected in many fits. The encoder-prepared and timestamped scores also agree only modestly. Source-video splits keep related clips together, but they do not ensure participant independence; the data lack person identifiers, a chronological split, and an external cohort, and the target and expanded readout were developed on this cohort. The score is not clinically validated. We verified aggregate arithmetic, but missing raw predictions and checkpoints prevented us from rerunning inference or the bootstrap. Together, these limits prevent us from claiming that S-JEPA training generally removes movement information.

For FMTS, this is an evaluation result rather than a claim about foundation-model scale. Temporal models should be tested on the quantities they are expected to support, not only on self-supervised objectives. Our masked task can use context from both sides of a hidden patch, so it does not test forecasting. A direct follow-up would predict future coordinates from a prefix in physical time and compare against last-position, constant-velocity, direct-pose, and initialized-encoder baselines. We have not evaluated forecasting or recursive rollout on real gait data. Here, S-JEPA training makes same-clip matching more consistent but reduces ridge-readout accuracy for the left-right speed score. The two metrics should be reported separately.

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164 A Implementation details and supplementary analyses

165 A.1 Data preparation

166 Of the 666 selected annotations, 642 had corresponding pose archives, and 625 clips passed the
167 quality checks. Each archive contains estimated landmark coordinates and visibility scores. The
168 coordinates are expressed relative to the video frame as $p_x = (x_0 + uW_c)/W$, $p_y = (y_0 + vH_c)/H$,
169 and $p_z = dW_c/W$. Here, (x_0, y_0) and (W_c, H_c) specify the crop origin and size, (W, H) is the size
170 of the full frame, and (u, v, d) is the pose estimate within the crop. These coordinates are image-
171 normalized estimates, including inferred depth, rather than calibrated physical measurements.

172 We center each pose at the midpoint of the hips and use a single body scale for each clip: the median
173 of the valid shoulder and hip widths. We divide the centered coordinates by this scale, using 1 if no
174 usable scale is available. A landmark is treated as observed only when its coordinates are finite and
175 its visibility score is at least 0.45. Computing the target requires both hips to be observed. The target
176 calculation retains the original observations and timestamps, without filling gaps or resampling.

177 The encoder input allows interpolation across gaps of at most four samples when observed endpoints
178 are available. If a pelvis midpoint is still unavailable, we use the clip’s median valid midpoint, or
179 zero if no valid midpoint exists. We resample the coordinates and validity mask to 64 positions. A
180 landmark remains usable only if its interpolated mask value is at least 0.999. Unusable coordinates
181 are set to zero. Each token represents four consecutive positions of one joint and is used only if all
182 four positions are valid. Thus, a $T_i \times 33 \times 4$ archive, with three coordinates and one visibility score
183 per landmark, produces 16×33 tokens with 96 features each.

184 A.2 Model training and masking

185 The encoder has four transformer layers and the predictor has two; both use 96-dimensional features
186 and four attention heads. We train with batches of 20 clips using AdamW, a learning rate of 0.001,
187 weight decay of 0.05, and a gradient-norm limit of 1. Training uses CUDA BF16, whereas evaluation
188 uses FP32. We update the teacher weights as an exponential moving average of the online encoder
189 weights, with coefficient 0.999.

190 The online encoder receives geometric views with rotations of up to 8 degrees and translations of
191 up to 0.03 in normalized coordinates. The teacher receives the prepared clip without these augmen-
192 tations. The cross-entropy loss on hidden features uses temperatures of 0.06 for the teacher and
193 0.10 for the predictor. Before forming the target probabilities, we center the teacher features using
194 a running mean with moving-average coefficient 0.9. The two-layer VICReg projector has width
195 96. Its invariance, variance, and covariance terms have weights 25, 25, and 1, respectively, and the
196 resulting regularization loss enters the training objective with weight 0.05.

197 For each motion masking condition, we count valid tokens among the twelve gait joints in each clip.
198 The number of hidden targets per clip is half the smallest count in the batch, rounded down, although
199 targets can be sampled from all 33 joints. MAMP-style sampling uses temperature 0.8. The mixture
200 condition combines 25% uniform sampling with 75% motion-weighted sampling; its motion scores
201 are capped at the 95th percentile of positive values before normalization.

202 A.3 Frozen readout and source weighting

203 For each of the five anatomical pairs, we summarize the left and right features using their means,
 204 standard deviations, and mean absolute changes between consecutive blocks. Means and standard
 205 deviations use only blocks in which both sides are valid; changes require both sides to be valid in
 206 both consecutive blocks. When a statistic has no valid supporting blocks or transitions, its value is
 207 set to zero. For each statistic, we retain the difference (left minus right) and the sum of the two sides,
 208 giving six 96-dimensional vectors per pair. We also record the fractions of blocks and transitions
 209 with valid measurements on both sides. Across five pairs, this gives $5 \times (6 \times 96 + 2) = 2,890$
 210 readout inputs. The candidate ridge penalties are 0.01, 0.1, 1, 10, 100, 1000, 10000.

211 Evaluation gives every source video equal total weight. For V videos, with n_v clips in video v , clip
 212 i receives weight $w_{vi} = 1/(Vn_v)$. We pool held-out predictions across all five outer folds within
 213 each seed and calculate

$$R_w^2 = 1 - \frac{\sum_{vi} w_{vi} (y_{vi} - \hat{y}_{vi})^2}{\sum_{vi} w_{vi} (y_{vi} - \bar{y}_w)^2}, \quad \bar{y}_w = \sum_{vi} w_{vi} y_{vi}.$$

214 The held-out mean $\bar{y}_w$ defines the evaluation statistic only; a deployable constant predictor would
 215 estimate its mean from training sources. The five test folds contain 19, 19, 19, 18, and 18 videos,
 216 corresponding to 189, 182, 72, 77, and 105 clips. We checked the assignments at the source-video
 217 level.

218 A.4 Supplementary analyses

219 The recorded mask samples show that motion masks hide approximately 17% of valid tokens and
 220 region masks hide 9.9%. Relative to random region masks, connected-region sampling reduces the
 221 proportion of targets with both immediate temporal neighbors visible from 69.9% to zero. Against
 222 their respective random controls, the MAMP-style, motion-mixture, and connected-region condi-
 223 tions yield changes in R^2 of -0.0008 , $+0.0005$, and -0.009 . The corresponding 95% intervals,
 224 obtained by resampling source videos, are $[-0.020, 0.015]$, $[-0.015, 0.015]$, and $[-0.033, 0.018]$.
 225 These exploratory intervals are not adjusted for multiple comparisons or choices made during devel-
 226 opment.

227 A 264-input baseline summarizing coordinates and validity reaches $R^2 = 0.035$. This result charac-
 228 terizes that particular summary and readout; it does not bound the information available in the poses.
 229 With the expanded readout, the largest tested ridge penalty, 10,000, is selected in 49/125 fits using
 230 trained teacher features and 96/125 fits using trained online encoder features. This concentration
 231 at the edge of the search range suggests that regularization warrants further study, but we have not
 232 established that larger penalties improve performance.

233 A.5 Interpretation and reproducibility

234 The dataset lacks persistent person identifiers and complete acquisition records. Clinical gait asym-
 235 metry motivates the speed score [8], but the score has not been clinically validated, and affected-side
 236 labels are unavailable. Averaging signed contrasts can also conceal differences between pairs when
 237 they favor opposite sides.

238 The retained evidence consists of aggregate scores and previously computed uncertainty intervals.
 239 Repeating encoder inference requires the model checkpoints and input data; repeating the source
 240 bootstrap requires the predictions for individual clips and their source assignments. The synthetic
 241 demonstrations exercise prediction from an observed prefix, but they do not establish forecasting or
 242 recursive rollout performance on real gait data. The supplement excludes videos and poses because
 243 access to the public annotations does not, by itself, authorize their redistribution.

244 **AI assistance.** An AI assistant supported evidence and literature checks, revisions to the text and
 245 figures, and critical review. The authors remain responsible for the final manuscript.